

Renaming, Substitution Congruence Standard Forms Prenex Forms

Logic in Computer Science Lab
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Based on prof. Tamás Kádek's lectures

Fixed Occurences

Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language, $A \in Form$ be a formula, and $x \in Var$ be a variable.

- A fixed occurrence of the variable x in the formula A is **free** if it is not in the subformulae $\forall xB$ or $\exists xB$ of the formula A .
- A fixed occurrence of the variable x in the formula A is **bound** if it is not free.

Open and Closed Formulae

Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language and $A \in Form$ be a formula.

- If $FreeVar(A) \neq \emptyset$, then the formula A is an **open formula**.
- If $FreeVar(A) = \emptyset$, then the formula A is a **closed formula**.

Remark

- The formula A is open if there is at least one variable which has at least one free occurrence in A .
- The formula A is closed if there is no variable which has a free occurrence in A .

Logical degree

The *logical degree* of a formula is a nonnegative integer defined as follows:

$$\ell : \textit{Form} \rightarrow \{0, 1, 2, \dots\}$$

If $A \in \textit{Con}$ then

- the logical degree of an atomic formula A is 0
 $\ell(A) \Rightarrow 0$;

if $B \in \textit{Form}$ and $C \in \textit{Form}$ are arbitrary formulas,

- $\ell(\neg B) \Rightarrow 1 + \ell(B)$;
- $\ell(B \circ C) \Rightarrow 1 + \ell(B) + \ell(C)$ where $\circ \in \{\wedge, \vee, \supset, \equiv\}$;
- $\ell(QxB) \Rightarrow 1 + \ell(B)$ where $Q \in \{\exists, \forall\}$ and $x \in \textit{Var}$.

Functional degree

The *functional degree* of a term is a nonnegative integer defined as follows:

$$\tilde{\ell} : \textit{Term} \rightarrow \{0, 1, 2, \dots\}$$

- If $t \in \mathcal{F}(0) \cup \textit{Var}$ then

$$\tilde{\ell}(t) \Rightarrow 0$$

(the functional degree of an atomic term t is 0)

- if $f(t_1, t_2, \dots, t_k) \in \textit{Term}$ then

$$\tilde{\ell}(t) \Rightarrow 1 + \sum_{i=1}^k \tilde{\ell}(t_i)$$

(the functional degree of a compound term is 1 plus the sum of the functional degrees of the direct subterms)

Free variable replacement

Bound variable renaming

Free variable replacement

Let $A \in \text{Form}$ be a formula, $x \in \text{Var}$ and $y \in \text{Var}$ be variables. The $[A_y^x]$ denotes the formula where all free occurrences of x is replaced by y .

Bound variable renaming

Renaming the bound variable x to y in the formula QxA has the result: $Qy[A_y^x]$.

The renaming is correct if

- $y \notin \text{FreeVar}(A)$, and
- there is no free occurrence of x in A which is in the scope of a quantifier bounding the variable y .

A renaming is correct, if y does not occur in the formula.

Replacement/Substitution

Formally, we define $F[t/x]$ by induction on terms and formulas as follows. On terms we have:

$$c[t/x] = c \quad \text{for } c \text{ a constant symbol}$$

$$y[t/x] = y \quad \text{for } y \neq x \text{ a variable}$$

$$x[t/x] = t$$

$$f(t_1, \dots, t_k)[t/x] = f(t_1[t/x], \dots, t_k[t/x]) \quad \text{for } f \text{ a } k\text{-ary function symbol}$$

We then extend the definition of $[t/x]$ to formulas as follows:

$$P(t_1, \dots, t_k)[t/x] = P(t_1[t/x], \dots, t_k[t/x])$$

$$(\neg F)[t/x] = \neg(F[t/x])$$

$$(F \wedge G)[t/x] = F[t/x] \wedge G[t/x]$$

$$(F \vee G)[t/x] = F[t/x] \vee G[t/x]$$

$$(Qy F)[t/x] = Qy (F[t/x]) \quad y \neq x \text{ a variable, } Q \in \{\forall, \exists\}$$

$$(Qx F)[t/x] = Qx F \quad Q \in \{\forall, \exists\}.$$

Replacement/Substitution

1. Decide whether or not the first variable is replaceable/substitutable with the second one. If it is replaceable, give the result of the substitution.

(b) $\forall x(\exists y P(y) \supset Q(x, y) \wedge Q(y, z))$

1. variable y is substitutable with variable x ?
2. variable y is substitutable with variable z ?
3. variable z is substitutable with variable x ?

(c) $\forall x Q(x, y) \supset \exists y R(x, y, z)$

1. variable x is substitutable with variable z ?
2. variable z is substitutable with variable x ?
3. variable z is substitutable with variable y ?

Solutions:

b) No, Yes, No

c) Yes, Yes, No

Replacement/Substitutions

Which are allowed substitutions where A is the formula

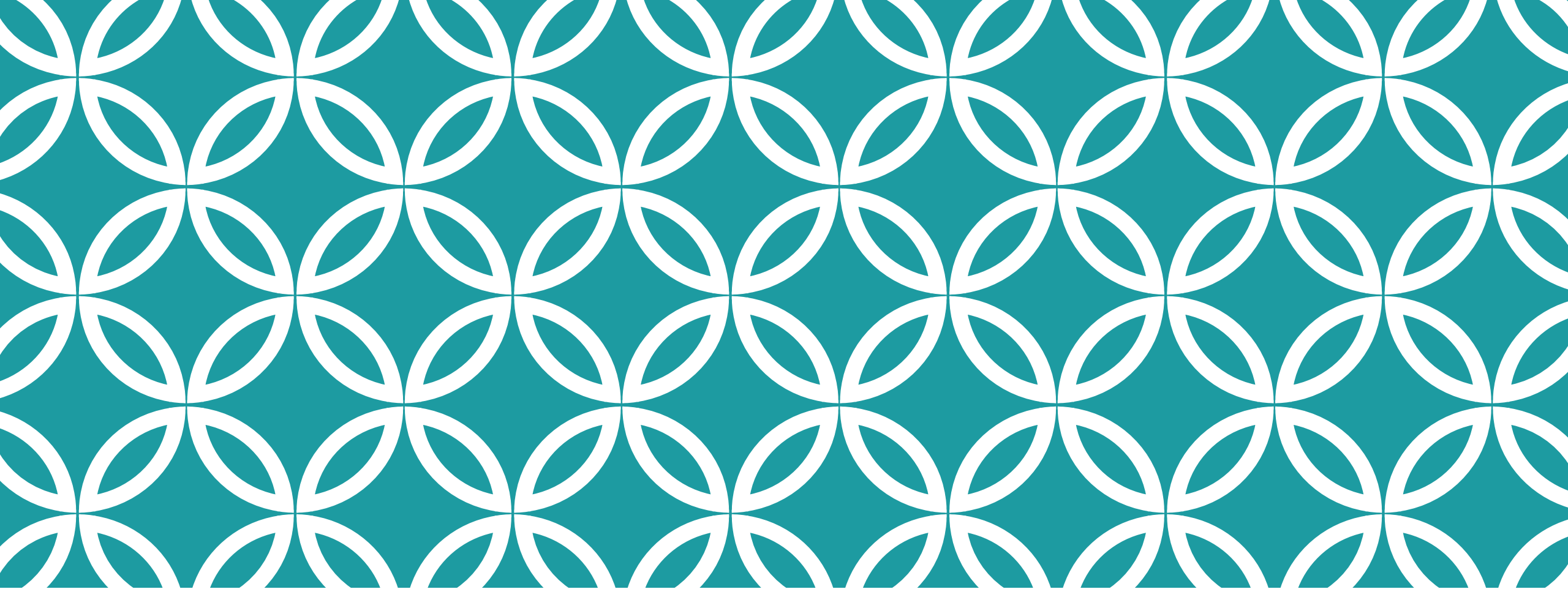
$$\neg(x = c \wedge Q(z, c)) \wedge \forall y(d = d \vee P(z))$$

a) $A(z||d)$

b) $A(z||x)$

c) $A(y||y)$

(A substitution is not allowed, if a free variable will be bound after the substitution)



Congruence |

Congruency

Skeleton of the formula:

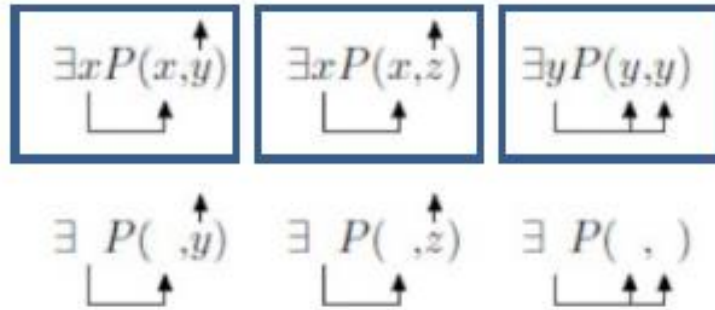
$$\forall x (A(x) \wedge \exists x B(x) \supset C(x)) \vee B(x)$$

$$\forall (A() \wedge \exists B()) \supset C()) \vee B(x)$$

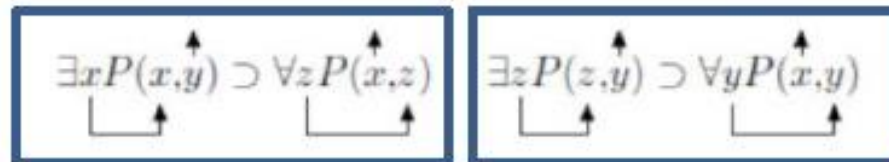
2 formulas are congruent if they have the same skeleton.

Congruence

(a) Not congruent formulae (not variants of themselves) because the



(b) Congruent formulae (variants)



Congruence

2. Perform regular renaming

In the formula $\forall x(\exists y P(y) \wedge \forall z Q(x, z)) \vee P(z)$ rename the sub-formula $\exists y A$ in a such way, that its quantor let be z !

Congruence

2. Perform regular renaming

In the formula $\forall x(\exists y P(y) \wedge \forall z Q(x, z)) \vee P(z)$ rename the sub-formula $\exists y A$ in a such way, that its quantor let be z !

Solution:

$$\forall x(\exists z P(z) \wedge \forall z Q(x, z)) \vee P(z)$$

Congruence

3. Perform regular renaming

In the formula $\forall x(\exists y P(y) \wedge \forall z Q(x, z)) \vee P(z)$ rename the sub-formula $\forall z A$ in a such way, that its quantor let be y !

Congruence

3. Perform regular renaming

In the formula $\forall x(\exists yP(y) \wedge \forall zQ(x, z)) \vee P(z)$ rename the sub-formula $\forall zA$ in a such way, that its quantor let be y !

Solution:

$$\forall x(\exists yP(y) \wedge \forall yQ(x, y)) \vee P(z)$$

Congruence

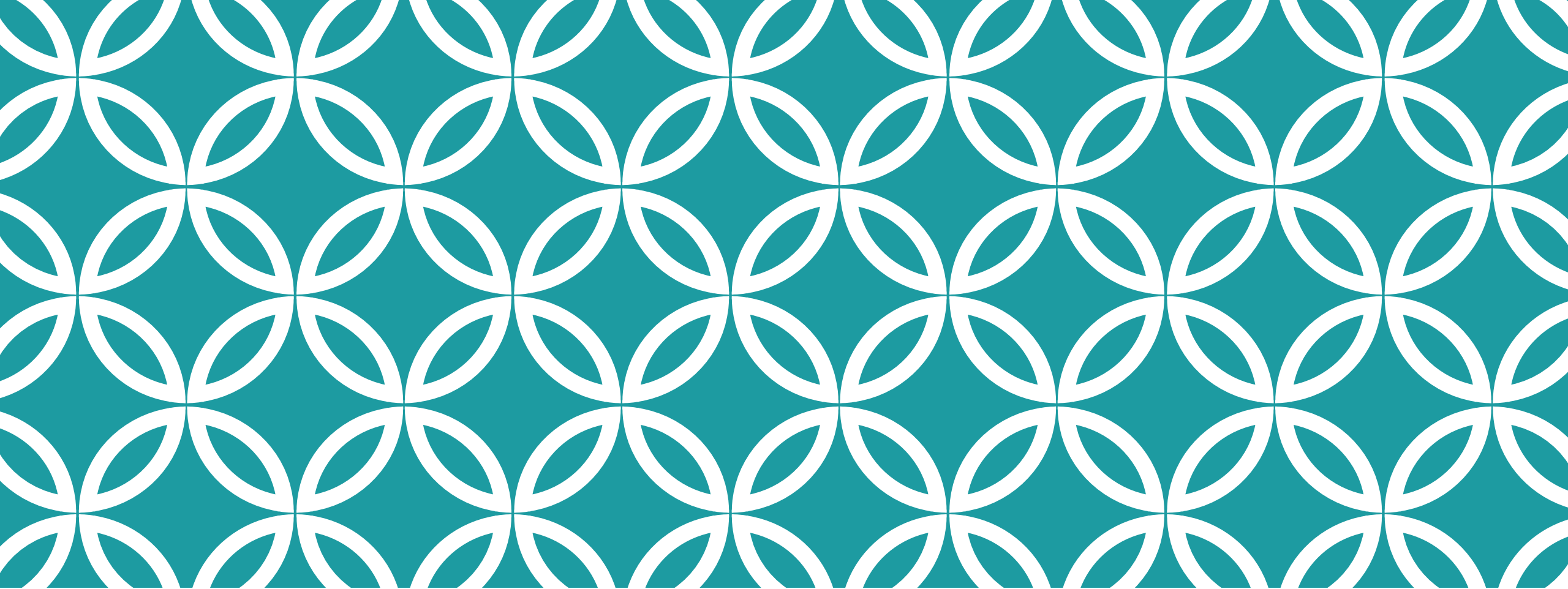
4. Decide whether or not the second formula is the regular renaming of the first one. Justify your answer.

(a) $\forall y(\exists x Q(y, x) \supset \forall y(Q(c, y) \wedge Q(y, x)))$ and $\forall z(\exists x Q(z, x) \supset \forall z(Q(c, z) \wedge Q(z, x)))$.

(b) $\forall y(\exists x Q(y, x) \supset \forall y(Q(c, y) \wedge Q(y, x)))$ and $\forall y(\exists x Q(y, x) \supset \forall z(Q(c, z) \wedge Q(z, x)))$.

(c) $\forall y(\exists x Q(y, x) \supset \forall y(Q(c, y) \wedge Q(y, x)))$ and $\forall z(\exists x Q(z, x) \supset \forall y(Q(c, y) \wedge Q(y, x)))$.

(d) $\forall x(\exists y Q(x, y) \supset \forall x(Q(c, x) \wedge Q(c, z)))$ and $\forall x(\exists y Q(x, y) \supset \forall z(Q(c, z) \wedge Q(c, z)))$.



Standard form



Standard form

A formula A is in *standard form* if

- the free and bound variables are different ($FreeVar(A) \cap BounVar(A) = \emptyset$),
- there is no two different quantifier bounding the same variable.

Create the standard forms of the formulae

1. $\exists z(P(y) \wedge \exists x P(f(x), z) \supset P(x, y))$

2. $\forall x(\exists z Q(x, z) \supset \forall z(Q(c, x) \wedge Q(x, y)))$

3. $\exists x \forall y Q(x, y) \supset \forall y \exists x Q(y, x)$

4. $\exists y \exists x \forall z((R(z, z) \supset y = z) \wedge \forall y(P(g(y, z)) \wedge y = x))$

4/b. $\exists y \exists x \forall z (R(z, z) \supset y = z) \wedge \forall y(P(g(y, z)) \wedge y = x)$

Standard form

1. Decide, whether the formulae below are standardized or not! If not, then construct its standardized form.

1. $\forall x(\exists yQ(x, y) \supset \forall x(Q(c, x) \wedge Q(c, z)))$

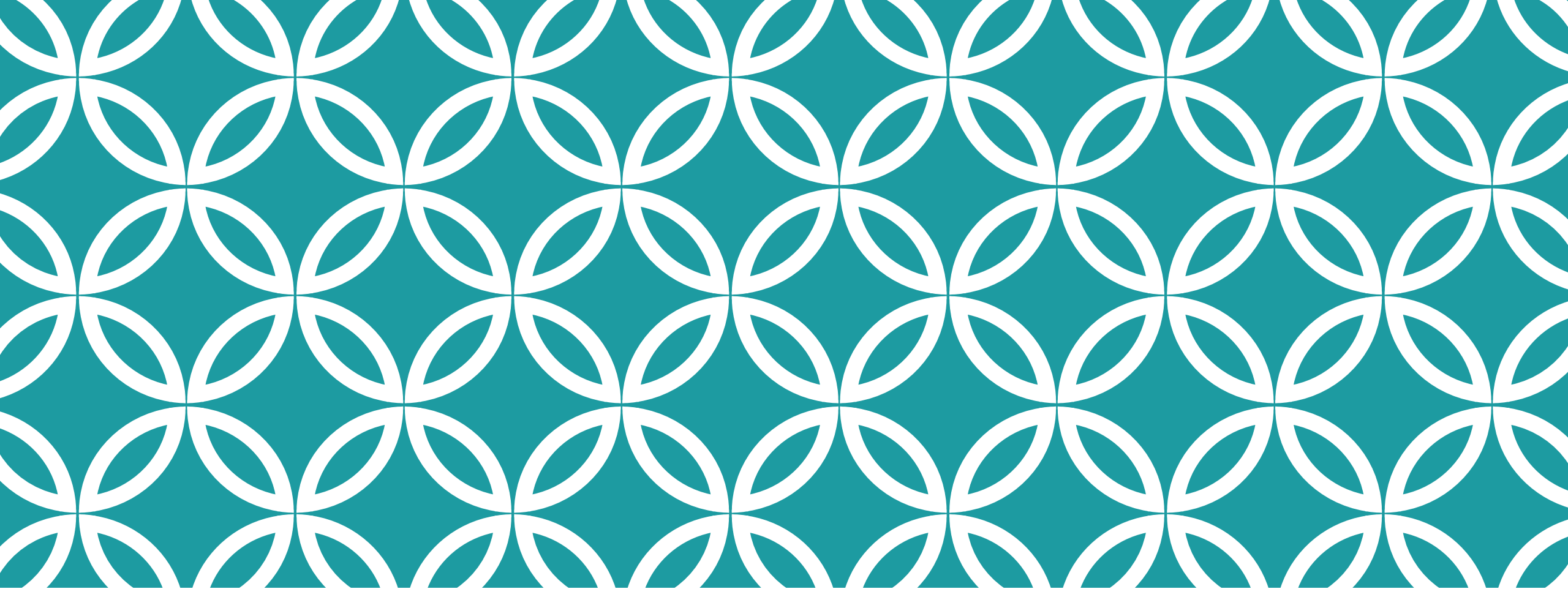
2. $\exists x\forall yQ(x, y) \supset \forall y\exists xQ(y, x)$

3. $\exists x(\forall xP(x) \supset Q(y, z) \wedge Q(y, x))$

4. $\exists x(P(y) \supset \exists yP(y, z) \vee \exists zP(z, y))$

5. $\exists z(P(y) \wedge \exists xP(x, z) \supset P(x, z))$

6. $\exists z(P(y) \wedge \exists xP(f(x), z) \supset P(x, y))$



Prenex Form |

Prenex form

Let $\langle LC, Var, Con, Term, Form \rangle$ be a first-order language. The formula

$$Q_1 x_1 Q_2 x_2 \dots Q_n x_n A$$

is in *prenex form* if A is a quantification free formula (A contains no quantifiers) and

- $Q_i \in \{\forall, \exists\}$ and,
- $x_i \in Var$

for all $i \in \{1, \dots, n\}$ where $n \geq 0$.

Prenex Form — How it looks like

$$\exists y(\forall x P(x) \supset \forall x Q(x, y))$$

The (original) form of the formula

$$\exists y(\forall x P(x) \supset \forall w Q(w, y))$$

A Standard Form of the formula

$$\exists y \exists x \forall w (P(x) \supset Q(w, y))$$

A Prenex Form of the formula

How to transform a formula to a Prenex Normal Form

1. Prepare the fully bracketed version of the formula
2. Construct a standard form of the formula
 - a) $\text{FreeVar}(B) \cap \text{BoundVar}(B) = \emptyset$
 - b) Rename variables so each variable following a quantifier has a unique name
3. Move $\neg$ “inwards” until there are no quantifiers in the scope of a negation, by de Morgan’s laws
4. Move quantifiers to the front of the sentence, without changing their order.

De Morgan's laws for quantifiers

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

De Morgan's laws

$$\forall x \neg P \equiv \neg \exists x P$$

$$\neg \forall x P \equiv \exists x \neg P$$

$$\forall x P \equiv \neg \exists x \neg P$$

$$\exists x P \equiv \neg \forall x \neg P$$

$$\neg P \wedge \neg Q \equiv \neg(P \vee Q)$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$P \wedge Q \equiv \neg(\neg P \vee \neg Q)$$

$$P \vee Q \equiv \neg(\neg P \wedge \neg Q)$$

Good to know

(A) De Morgan's laws for quantifiers

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

(B) If x does not occur free in G then:

$$(\forall x F \wedge G) \equiv \forall x (F \wedge G)$$

$$(\forall x F \vee G) \equiv \forall x (F \vee G)$$

$$(\exists x F \wedge G) \equiv \exists x (F \wedge G)$$

$$(\exists x F \vee G) \equiv \exists x (F \vee G)$$

$$(C) (\forall x F \wedge \forall x G) \equiv \forall x (F \wedge G)$$

$$(\exists x F \vee \exists x G) \equiv \exists x (F \vee G)$$

$$(D) \forall x \forall y F \equiv \forall y \forall x F$$

$$\exists x \exists y F \equiv \exists y \exists x F$$

$$(E) \forall x F \supset G \equiv \exists x (F \supset G)$$

$$\exists x F \supset G \equiv \forall x (F \supset G)$$

$$F \supset \forall x G \equiv \forall x (F \supset G)$$

$$F \supset \exists x G \equiv \exists x (F \supset G)$$

Prenex form

Construct the prenex form of the following formulae

a) $\forall x P(x) \supset \neg \exists x P(x) \vee Q(x, c)$

Solution:

Standard form

A formula A is in *standard form* if

- the free and bound variables are different ($\text{FreeVar}(A) \cap \text{BoundVar}(A) = \emptyset$),
- there is no two different quantifier bounding the same variable.

$$\neg \forall x P(x) \Leftrightarrow \exists x \neg P(x)$$

$$\neg \exists x P(x) \Leftrightarrow \forall x \neg P(x)$$

$$\neg P \wedge \neg Q \equiv \neg(P \vee Q)$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \supset \forall x B \Leftrightarrow \forall x (A \supset B)$$

$$2. \forall x B \supset A \Leftrightarrow \exists x (B \supset A)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \supset \exists x B \Leftrightarrow \exists x (A \supset B)$$

$$2. \exists x B \supset A \Leftrightarrow \forall x (B \supset A)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \wedge \forall x B \Leftrightarrow \forall x (A \wedge B)$$

$$2. A \wedge \exists x B \Leftrightarrow \exists x (A \wedge B)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \vee \forall x B \Leftrightarrow \forall x (A \vee B)$$

$$2. A \vee \exists x B \Leftrightarrow \exists x (A \vee B)$$

Construct the prenex form of the following formula

1. $\forall x(\exists y Q(x, y) \supset \forall x(Q(c, x) \wedge Q(c, z)))$

$$\forall w \forall x \forall y (Q(w, y) \supset (Q(c, x) \wedge Q(c, z))).$$

Prenex form

Construct the prenex form of the following formula

$$\text{b) } \exists x \forall y P(x,y) \vee \exists x \forall y Q(x,y)$$

Solution:

$$\exists x \forall y \exists z \forall u (P(x,y) \vee Q(z,u))$$

$$\neg \forall x P(x) \Leftrightarrow \exists x \neg P(x)$$

$$\neg \exists x P(x) \Leftrightarrow \forall x \neg P(x)$$

$$\neg P \wedge \neg Q \equiv \neg(P \vee Q)$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \supset \forall x B \Leftrightarrow \forall x (A \supset B)$$

$$2. \forall x B \supset A \Leftrightarrow \exists x (B \supset A)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \supset \exists x B \Leftrightarrow \exists x (A \supset B)$$

$$2. \exists x B \supset A \Leftrightarrow \forall x (B \supset A)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \wedge \forall x B \Leftrightarrow \forall x (A \wedge B)$$

$$2. A \wedge \exists x B \Leftrightarrow \exists x (A \wedge B)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \vee \forall x B \Leftrightarrow \forall x (A \vee B)$$

$$2. A \vee \exists x B \Leftrightarrow \exists x (A \vee B)$$

Prenex form

Construct the prenex form of the following formula

$$h) \quad \forall x(\exists y Q(x, y) \supset \forall x P(x)) \supset \neg(\forall x P(x) \vee \forall x R(x))$$

Solution:

$$\exists x \exists y \exists z \exists u \exists w ((Q(x, y) \supset P(z)) \supset \neg(P(u) \vee R(w)))$$

$$\neg \forall x P(x) \Leftrightarrow \exists x \neg P(x)$$

$$\neg \exists x P(x) \Leftrightarrow \forall x \neg P(x)$$

$$\neg P \wedge \neg Q \equiv \neg(P \vee Q)$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \supset \forall x B \Leftrightarrow \forall x (A \supset B)$$

$$2. \forall x B \supset A \Leftrightarrow \exists x (B \supset A)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \supset \exists x B \Leftrightarrow \exists x (A \supset B)$$

$$2. \exists x B \supset A \Leftrightarrow \forall x (B \supset A)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \wedge \forall x B \Leftrightarrow \forall x (A \wedge B)$$

$$2. A \wedge \exists x B \Leftrightarrow \exists x (A \wedge B)$$

If $x \notin \text{FreeVar}(A)$, then

$$1. A \vee \forall x B \Leftrightarrow \forall x (A \vee B)$$

$$2. A \vee \exists x B \Leftrightarrow \exists x (A \vee B)$$

Prenex form

Mark the prenex form of formula $P(d) \vee \exists x \forall x \exists z ((Q(x, z) \supset R(z, y)) \wedge \exists y \exists x y = c)$, where $F(0)=\{c\}$, $F(1)=\{f\}$, $F(2)=\{g, h\}$, $P(1)=\{P\}$ and $P(2)=\{Q, R\}$?

Select one or more:

- ☐ a. $\forall w \exists x \exists u \exists z (P(d) \vee (Q(w, z) \supset R(z, y)) \wedge u = c)$
-  ☐ b. $\forall w \exists u \exists z \exists v (P(d) \vee (Q(w, z) \supset R(z, y)) \wedge u = c)$
- ☐ c. $\exists u \exists z \exists x \forall w (P(d) \vee (Q(w, z) \supset R(z, y)) \wedge u = c)$
- ☐ d. $\exists x \exists z \exists v \forall w \exists u (P(d) \vee (Q(w, z) \supset R(z, y)) \wedge u = c)$

Prenex form

Mark the prenex form of formula $\exists z \exists y \exists x (\exists x (R(x, z) \supset P(x) \wedge R(d, h(c, d))) \wedge \exists y d = g(z, z))$, where $F(0) = \{c\}$, $F(1) = \{f\}$, $F(2) = \{g, h\}$, $P(1) = \{P\}$ and $P(2) = \{Q, R\}$?

Select one or more:

-  ☐ a. $\exists u \exists z \exists x \exists y ((R(u, z) \supset P(u) \wedge R(d, h(c, d))) \wedge d = g(z, z))$
-  ☐ b. $\exists u \exists z ((R(u, z) \supset P(u) \wedge R(d, h(c, d))) \wedge d = g(z, z))$
-  ☐ c. $\exists u \exists v \exists z \exists x ((R(u, z) \supset P(u) \wedge R(d, h(c, d))) \wedge d = g(z, z))$
-  ☐ d. $\exists z \exists y \exists u ((R(u, z) \supset P(u) \wedge R(d, h(c, d))) \wedge d = g(z, z))$

Prenex form

Mark the prenex form of formula $\neg(x = x \wedge \neg d = f(x)) \wedge \exists z(Q(d, x) \supset \forall z \forall x R(z, f(c)))$, where $F(0)=\{c\}$, $F(1)=\{f\}$, $F(2)=\{g, h\}$, $P(1)=\{P\}$ and $P(2)=\{Q, R\}$?

Select one or more:

- ☐ a. $\exists z \forall v (\neg(x = x \wedge \neg d = f(x)) \wedge (Q(d, x) \supset R(v, f(c))))$
-  ☐ b. $\forall u \exists z \forall v (\neg(x = x \wedge \neg d = f(x)) \wedge (Q(d, x) \supset R(v, f(c))))$
- ☐ c. $\exists z \forall v \forall u (\neg(x = x \wedge \neg d = f(x)) \wedge (Q(d, x) \supset R(v, f(c))))$
- ☐ d. $\forall v \exists z (\neg(x = x \wedge \neg d = f(x)) \wedge (Q(d, x) \supset R(v, f(c))))$

Prenex form

Mark the prenex form of formula $Q(x, y) \supset R(c, z) \supset R(z, c) \vee \forall x \forall y \neg \forall y (c = d \supset c = z)$, where $F(0)=\{c\}$, $F(1)=\{f\}$, $F(2)=\{g, h\}$, $P(1)=\{P\}$ and $P(2)=\{Q, R\}$?

Select one or more:

- ☐ a. $\forall v (Q(x, y) \supset R(c, z) \supset R(z, c) \vee \neg (c = d \supset c = z))$
- ☐ b. $\forall u \forall w (Q(x, y) \supset R(c, z) \supset R(z, c) \vee \neg (c = d \supset c = z))$
- ☐ c. $\forall v \forall w (Q(x, y) \supset R(c, z) \supset R(z, c) \vee \neg (c = d \supset c = z))$
- ☐ d. $\forall v \forall u (Q(x, y) \supset R(c, z) \supset R(z, c) \vee \neg (c = d \supset c = z))$

Which is/are prenex form(s) of the formula?

Mark the prenex form of formula

$x = d \supset \exists z \exists y \forall y \forall x (Q(c, c) \vee Q(x, z) \wedge \forall x (c = x \supset z = d))$, where $F(0)=\{c\}$, $F(1)=\{f\}$, $F(2)=\{g,h\}$, $P(1)=\{P\}$ and $P(2)=\{Q,R\}$?

Select one or more:

- ☐ a. $\exists y \forall v \exists z \forall u (x = d \supset Q(c, c) \vee Q(v, z) \wedge (c = u \supset z = d))$
-  ☐ b. $\exists z \forall u \forall v \forall w (x = d \supset Q(c, c) \vee Q(v, z) \wedge (c = u \supset z = d))$
- ☐ c. $\exists y \forall w \exists z \forall u \forall v (x = d \supset Q(c, c) \vee Q(v, z) \wedge (c = u \supset z = d))$
- ☐ d. $\forall u \exists z \exists y \forall v (x = d \supset Q(c, c) \vee Q(v, z) \wedge (c = u \supset z = d))$

Which is/are prenex form(s) of the formula?

Mark the prenex form of formula $\exists y \exists x \forall z ((R(z, z) \supset y = z) \wedge \forall y (P(g(y, z)) \wedge y = x))$, where $F(0) = \{c\}$, $F(1) = \{f\}$, $F(2) = \{g, h\}$, $P(1) = \{P\}$ and $P(2) = \{Q, R\}$?

Select one or more:

- ☐ a. $\exists y \forall u \forall z \exists x ((R(z, z) \supset y = z) \wedge P(g(u, z)) \wedge u = x)$
- ☐ b. $\exists x \forall u \exists y \forall z ((R(z, z) \supset y = z) \wedge P(g(u, z)) \wedge u = x)$
- ☐ c. $\forall u \forall z \exists y \exists x ((R(z, z) \supset y = z) \wedge P(g(u, z)) \wedge u = x)$
-  ☐ d. $\exists y \exists x \forall z \forall u ((R(z, z) \supset y = z) \wedge P(g(u, z)) \wedge u = x)$